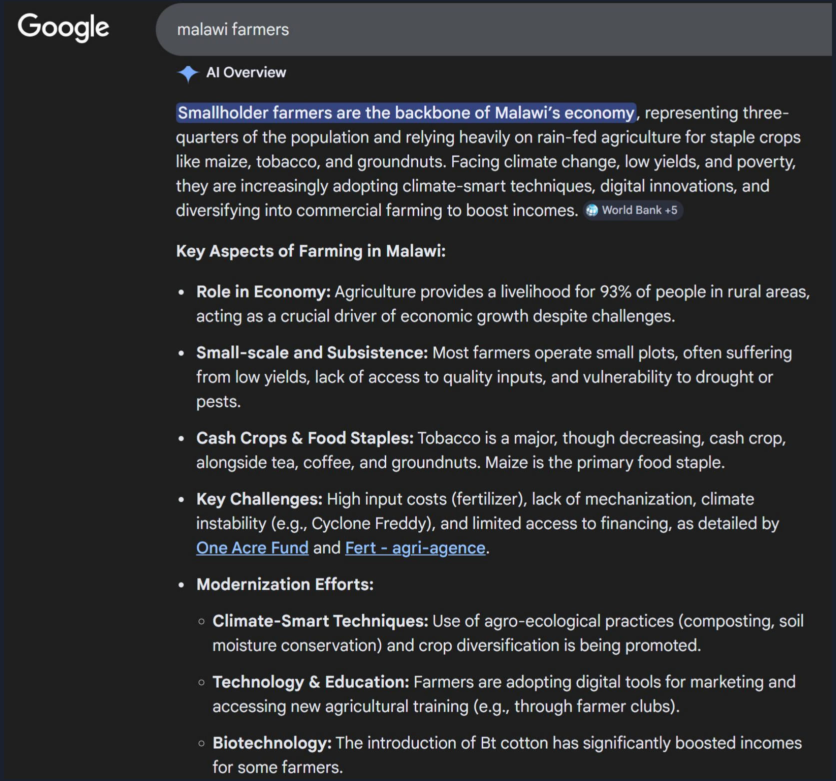
# Midterm presentation

Group 7 : Luis Casado, Antoine De Vinck, Alex Sanmartin, Eric Pizem, Jonas Harbarth

# Project motivation and opportunity gap



Google

malawi farmers

AI Overview

Smallholder farmers are the backbone of Malawi's economy, representing three-quarters of the population and relying heavily on rain-fed agriculture for staple crops like maize, tobacco, and groundnuts. Facing climate change, low yields, and poverty, they are increasingly adopting climate-smart techniques, digital innovations, and diversifying into commercial farming to boost incomes. World Bank +5

**Key Aspects of Farming in Malawi:**

- **Role in Economy:** Agriculture provides a livelihood for 93% of people in rural areas, acting as a crucial driver of economic growth despite challenges.
- **Small-scale and Subsistence:** Most farmers operate small plots, often suffering from low yields, lack of access to quality inputs, and vulnerability to drought or pests.
- **Cash Crops & Food Staples:** Tobacco is a major, though decreasing, cash crop, alongside tea, coffee, and groundnuts. Maize is the primary food staple.
- **Key Challenges:** High input costs (fertilizer), lack of mechanization, climate instability (e.g., Cyclone Freddy), and limited access to financing, as detailed by One Acre Fund and Fert - agri-agence.
- **Modernization Efforts:**
  - **Climate-Smart Techniques:** Use of agro-ecological practices (composting, soil moisture conservation) and crop diversification is being promoted.
  - **Technology & Education:** Farmers are adopting digital tools for marketing and accessing new agricultural training (e.g., through farmer clubs).
  - **Biotechnology:** The introduction of Bt cotton has significantly boosted incomes for some farmers.

Major constraints :

- Low cost
- Constant spacing
- Planting on ridges

# Target User and Scenario, Mission Statement

- **Who is the user?**
  Farmers growing wheat, corn or other cereal crops.
- **What do they need?**
  They need to sow seeds and fertilizer efficiently and evenly over large areas.
- **Why do they need it?**
  They need it to spread the seeds/fertilizer in a regular and uniform way.
- **How is the task or activity currently done?**
  The farmer plant the seeds by hand.
- **Where does it take place?**
  In their agricultural fields, on soil that has been prepared after plowing.
- **When does it take place?**
  At the optimal time for each crop, based on seasonal and weather conditions.

# Target User and Scenario, Mission Statement

| Product Opportunity Gap | Hand-operated precision planter and fertilizer applier that is durable and has an accessible price for the farmers |
|---|---|
| Benefit Proposition | • Ensures correct seed planting depth and consistent spacing.<br>• Place fertilizer safely beside seeds.<br>• Reduces physical strain from bending and repetitive digging.<br>• Improves germination rates and reduces replanting. |
| Key Business Goals | • Design a low-cost manual precision seeder, easy to use and durable.<br>• Develop a functional prototype within the project timeline. |
| Target Market | • Smallholder farmers in Malawi relying primarily on manual agricultural tools. |
| Secondary Market | • Smallholders farmers in other countries with similar soil conditions.<br>• NGOs supporting rural farming.<br>• Garden owners. |
| Assumptions and Constraints | • Fully manual operation (no fuel or electricity).<br>• Must function in ridge-based planting systems.<br>• Must operate effectively in heavy or sticky clay soils.<br>• Must remain affordable and easy to repair locally. |
| Stakeholders | • Farmers.<br>• Manufacturers and distributors.<br>• Designers and engineers involved in the project. |

# Target Product Specifications

*Importance 5*

| Metric No. | Need Nos. | Metric | Importance | Units | Marginal Value | Ideal Value |
|---|---|---|---|---|---|---|
| 1 | 1 | Planting speed | 5 | ha/day | ≥2 | ≥4 |
| 2 | 2 | Seed coverage rate | 5 | % | ≥85 | ≥95 |
| 3 | 3 | Distance between planting stations | 5 | cm | 20-30 | 25 |
| 4 | 3 | Number of seeds placed per station | 5 | count | 1–3 ±1 | Selectable {1;2;3} ±0 |
| 5 | 3 | Distance between seed and fertilizer | 5 | cm | 4-12 | 5–10 |
| 6 | 4 | Hole depth created | 5 | cm | 5–30 adjustable | 5–20 adjustable (±1 cm control) |
| 7 | 7 | Number of bending motions required per h | 5 | count | ≤60 | ≤10 |
| 8 | 10 | External energy required | 5 | binary | None | None |
| 9 | 19 | hours to failure in harsh soil | 5 | hours | ≥30,000 | ≥100,000 |
| 10 | 20 | Target unit retail cost | 5 | USD | ≤80 | ≤50 |
| 11 | 3 | Planting accuracy (depth & spacing) | 5 | % | ≥80 | ≥95 |

# Target Product Specifications

*Importance 4*

| Metric No. | Need Nos. | Metric | Importance | Units | Marginal Value | Ideal Value |
|---|---|---|---|---|---|---|
| 12 | 3 | Mass of fertilizer dispensed per station | 4 | g | 2-6 | 3-5 |
| 13 | 7 | Total mass of empty system | 4 | kg | ≤10 | ≤6 |
| 14 | 9 | Number of operators required | 4 | persons | ≤2 | 1 |
| 15 | 14 | Ground pressure on ridges | 4 | kPa | ≤30 | ≤15 |
| 16 | 18 | Number of moving parts | 4 | count | ≤25 | ≤15 |
| 17 | 18 | Soil on PSF after 1h of use | 4 | g | ≤150 | ≤50 |
| 18 | 20 | Replacement parts cost per year | 4 | USD | ≤3 | ≤1 |

# Target Product Specifications

*Importance 3 & 2*

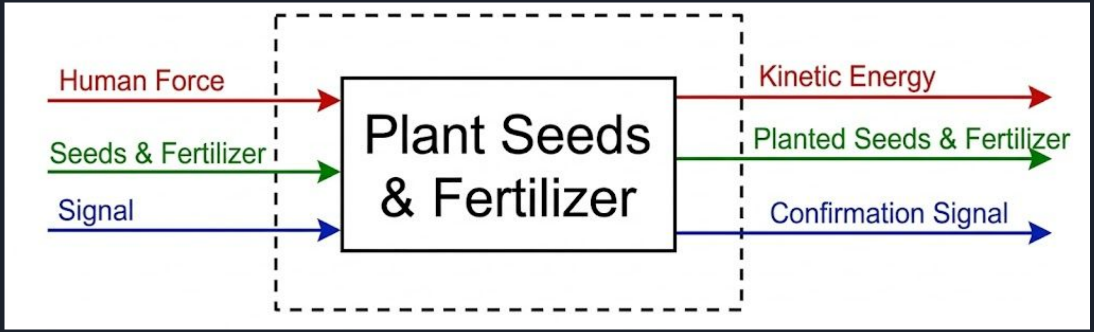
| Metric No. | Need Nos. | Metric | Importance | Units | Marginal Value | Ideal Value |
|---|---|---|---|---|---|---|
| 19 | 5 | Seed hopper max capacity | 3 | kg | ≥1 | ≥2 |
| 20 | 16 | Transport dimensions (L×W×H) | 3 | cm | ≤120×60×60 | ≤100×50×50 |
| 21 | 17 | TIME required for cleaning | 3 | Min /day of use | ≤30 | ≤20 |
| 22 | 18 | Locally available components | 3 | % | ≥70 | ≥90 |
| 23 | 1 | Overall planting precision at target speed | 3 | % | ≥80 | ≥95 |
| 24 | 8 | Storage volume required | 2 | $dm^3$ | ≤120 | ≤60 |
| 25 | 13 | Adjustable planting spacing range | 2 | cm | 50–90 | 25–100 |

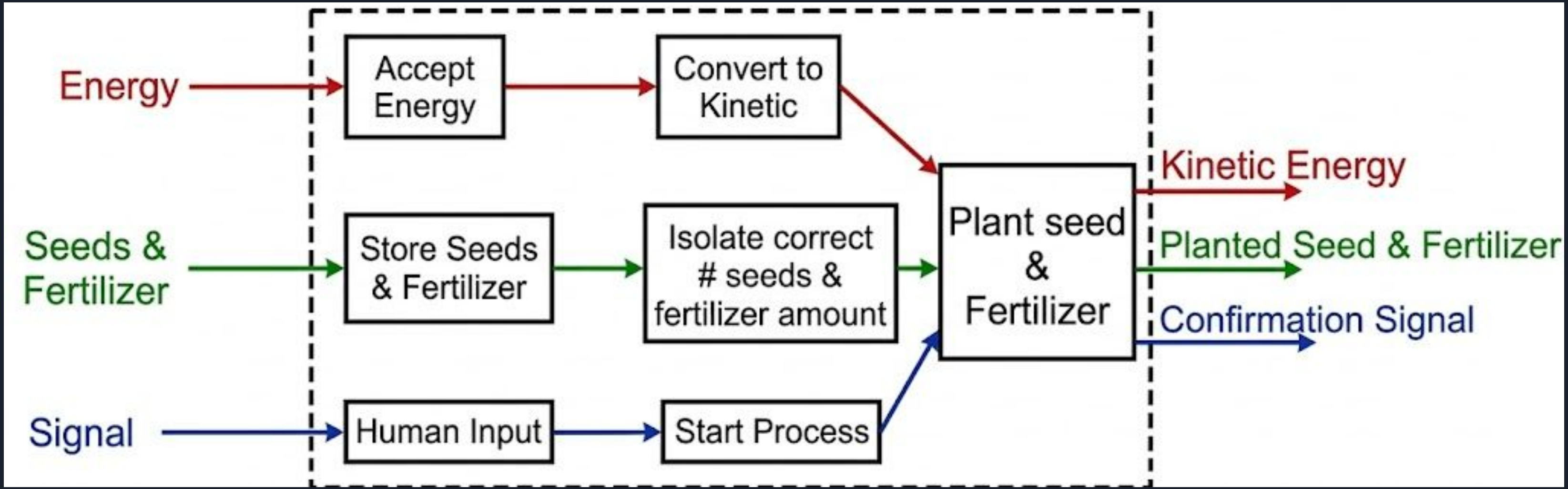
# Methods for concept generation

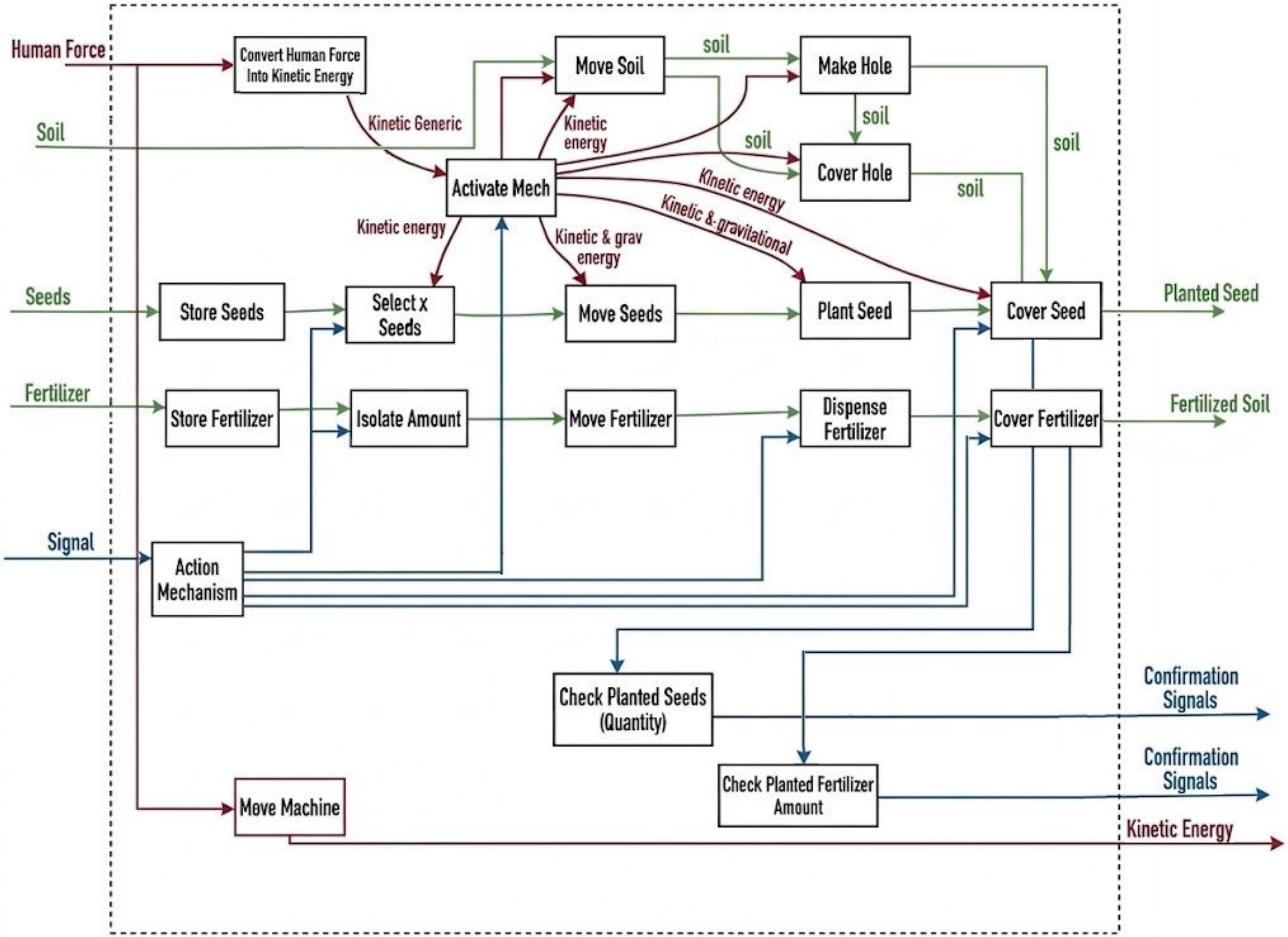
- First method: Functional Decomposition
  - Define the fundamental product tasks.
  - Decompose overall system into subsystems.
  - Make combination of components to generate different concepts.

- Second method: Brainwriting
  - Format 3-2-15 → 6 concepts in total.
  - 3 members, 2 concepts each, rotation every 15 minutes.
  - Heuristic Cards are applied during the design and after.

# Functional Decomposition



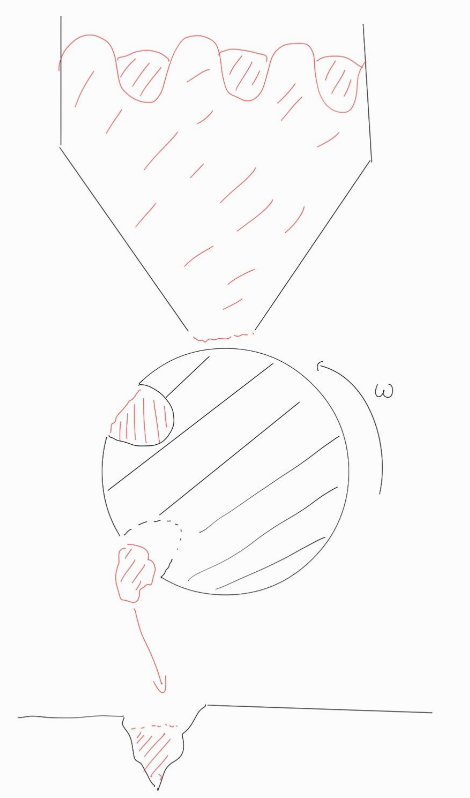

Human Force
Convert Human Force Into Kinetic Energy
Soil
Kinetic Generic
Move Soil
soil
Make Hole
Kinetic energy
soil
soil
Cover Hole
soil
soil
Activate Mech
Kinetic energy
Kinetic & gravitational
Kinetic energy
Kinetic & grav energy
Seeds
Store Seeds
Select x Seeds
Move Seeds
Plant Seed
Cover Seed
Planted Seed
Fertilizer
Store Fertilizer
Isolate Amount
Move Fertilizer
Dispense Fertilizer
Cover Fertilizer
Fertilized Soil
Signal
Action Mechanism
Check Planted Seeds (Quantity)
Confirmation Signals
Check Planted Fertilizer Amount
Confirmation Signals
Move Machine
Kinetic Energy

# Concept Combination Table

| Store Seeds and Fertilizer | Isolate Correct amount | Deliver the Load |
|---|---|---|
| Hopper | Empty Drum with exact volume | Gravity |
| Bucket | Infinite Screw | Articulated Claw |
| Bags | Infinite Zig-Zag Saw | Push System |

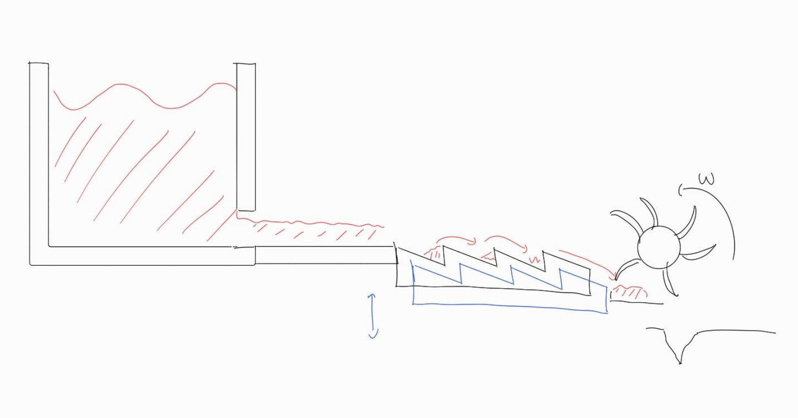
# Concept Combination Table

| Store Seeds and Fertilizer | Isolate Correct amount | Deliver the Load |
|---|---|---|
| Hopper | Empty Drum with exact volume | Gravity |
| Bucket | Infinite Screw | Articulated Claw |
| Bags | Infinite Zig-Zag Saw | Push System |



# Concept Combination Table

| Store Seeds and Fertilizer | Isolate Correct amount | Deliver the Load |
|---|---|---|
| Hopper | Empty Drum with exact volume | Gravity |
| Bucket | Infinite Screw | Articulated Claw |
| Bags | Infinite Zig-Zag Saw | Push System |

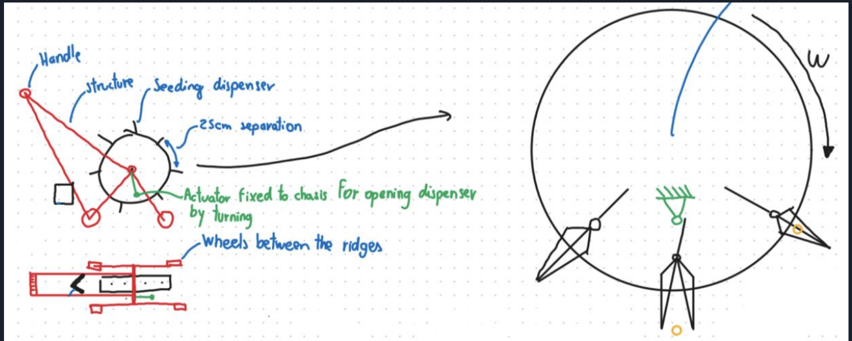


# Concept Combination Table

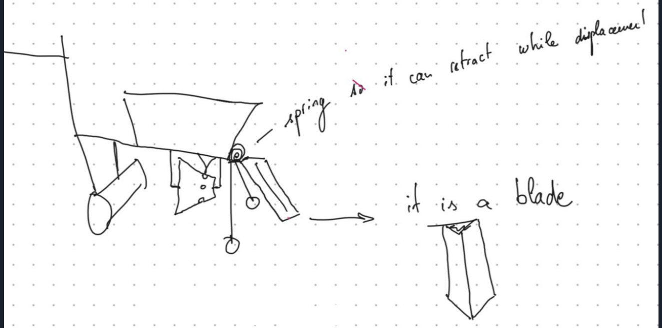
| Store Seeds and Fertilizer | Isolate Correct amount | Deliver the Load |
|---|---|---|
| Hopper | Empty Drum with exact volume | Gravity |
| Bucket | Infinite Screw | Articulated Claw |
| Bags | Infinite Zig-Zag Saw | Push System |

# Three best concepts

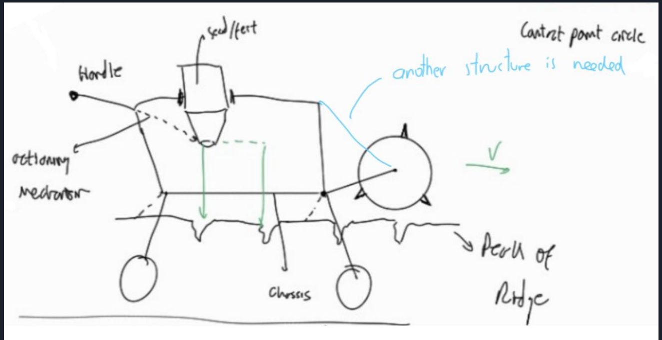
1.




3.




2.



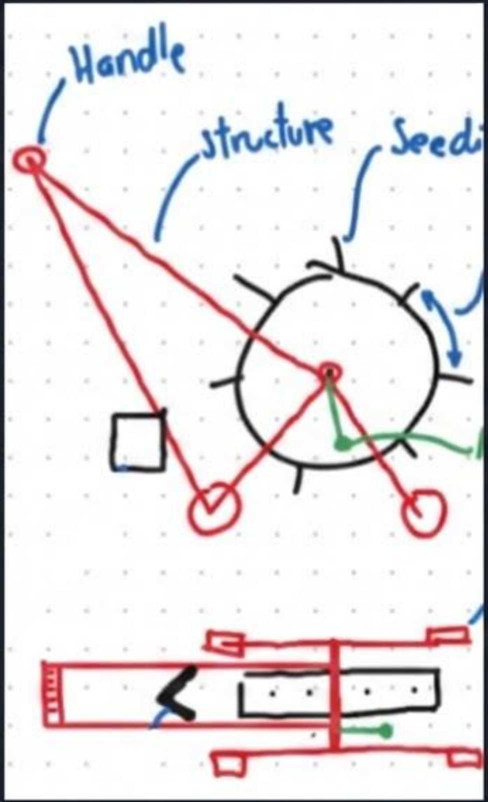
# Selection Method

After brainstorming, we selected the six best concepts to apply the Pugh Concept Screening Method (decision matrix) and identify the three most promising concepts.

| | CONCEPT VARIANTS | | | | | | |
|---|---|---|---|---|---|---|---|
| **Selection Criteria** | **A** | **B** | **C** | **D** | **E** | **F** | **REF** |
| Ease of Operation | - | 0 | + | 0 | - | - | 0 |
| Number of Pieces | 0 | + | - | 0 | - | - | 0 |
| Weight | 0 | - | - | 0 | - | - | 0 |
| Manufacturing Ease | + | + | 0 | + | - | 0 | 0 |
| Portability | + | + | + | 0 | + | + | 0 |
| Ridge Conservation | + | 0 | + | 0 | + | + | 0 |
| Isolation of Seeds | - | 0 | 0 | - | - | + | 0 |
| Consistent spacing | - | - | 0 | 0 | - | - | 0 |
| Hole coverage | - | 0 | + | 0 | - | - | 0 |
| Hole Creation | 0 | - | 0 | 0 | 0 | + | 0 |
| Maintenance and Repair | + | 0 | 0 | - | - | + | 0 |
| Human Energy | - | 0 | 0 | - | - | - | 0 |
| Cost | + | - | 0 | - | - | - | 0 |
| Variable setups | + | + | 0 | 0 | 0 | + | 0 |
| PLUSES | 6 | 4 | 4 | 1 | 2 | 6 | |
| SAMES | 3 | 6 | 8 | 9 | 2 | 1 | |
| MINUSES | 5 | 4 | 2 | 4 | 10 | 7 | |
| | 14 | 14 | 14 | 14 | 14 | 14 | |
| NET | 1 | 0 | 2 | -3 | -8 | -1 | |
| RANK | 2 | 3 | 1 | 5 | 6 | 4 | |
| CONTINUE? | YES | NO | YES | NO | NO | NO | |

Reference :

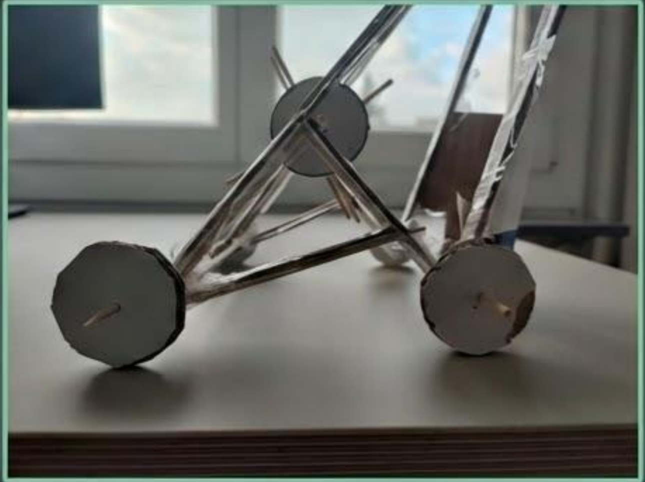
# First Prototype Concept (C)




# First Prototype Concept

The mechanism

the structure



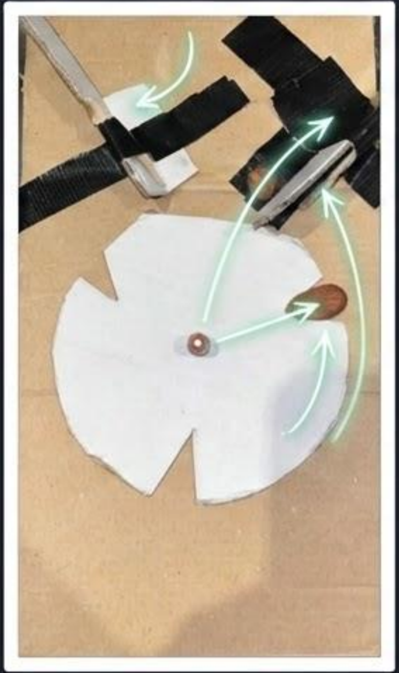
# First Prototype Concept

The blade

- ***The main innovation*** *adopted into the final concept*

# Second Prototype Concept (A)

the selection mechanism



- customizable
- precise selection every 25 cm

# Next steps

- Detailed Design Phase
- Which part to make? Which to buy?
- Material : Durability? Low cost? Repairability?

# Thanks for your attention !

QUESTIONS ?